www.csvtuonline.com

Printed Pages – 7

29/4/14

337811 (37)

525

BE (8th Semester)

Examination, April-May, 2014

Branch : Mechanical

ROBOTICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

REFERENCE BOOK

Note : Attempt all questions. Part (a) of each question is compulsory. Attmept any two parts from (b), (c) and (d).

UNIT - I

Q. 1. (a) Name the four basic components of a robot system. 2

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(b) What are the advantages and disadvantages of cylindrical arm configuration over a polar arm configuration. Explain both configurations in detail. 7

(c) Explore the anatomy of the human wrist joint and analyze it for type of motions provided, number of degrees of freedom, number of joints, type of joints etc. 7

(d) Describe the working of some contact sensors used in robotics. Give advantages and disadvantages of each. 7

www.csvtuonline.com

www.csvtuonline.com

UNIT - II

Q. 2. (a) What do you mean by mapping ? 2

(b) Show that the determinant of the rotation matrix R, is +1 for a right-hand coordinate system and –1 for a left hand coordinate system. 7

(c) Determine the transformation matrix T that represents a translation of a unit along x-axis, followed by a rotation of angle α about x-axis followed by a rotation of θ about the rotated z-axis. 7

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(d) Determine the rotation matrix for a rotation of 45° about y-axis, followed by a rotation of 120° about z-axis, and a final rotation of 90° about x-axis. 7

UNIT - III

Q. 3. (a) Draw a 3-DOF manipulator arm and a show numbering of links and joints. 2

(b) Obtain the direct kinematics equation of the 4-DOF Selective Compliance Assembly Robot Arm (SCARA) robots. 14

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

Or

Draw a five degree of freedom industrial manipulator and obtain the transformation matrices for it's different components and overall transformation matrix for the manipulator. 14

UNIT - IV

Q. 4. (a) Define quantization error ? 2

(b) Discuss the industrial and non industrial applications of vision-controlled robots. 7

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(c) Show how image storage requirements depend on spatial and amplitude digitization parameters. 7

(d) How robotic sensors are classified ? Give brief description of their working and use. 7

UNIT - V

Q. 5. (a) Are robots super workers ? 2

(b) What are the basic principles for robot applications in industries. What are the stages of robot application planning for industrial applications. 7

www.csvtuonline.com

www.csvtuonline.com

(c) Describe the term "Robot Safety". Why robots are useful in industries ? 7

(d) Explain the types of manipulator which is best suited for machine loading and unloading application and assembly application. 7

www.csvtuonline.com

www.csvtuonline.com